

JIAJUN ZHU

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EDUCATION

The University of Texas at Austin

Ph.D. in Electrical and Computer Engineering

Sept. 2025 - Present

Zhejiang University

B.S. in Mathematics

Sept. 2020 - Jun. 2025

Major GPA: 3.9/4.0

PUBLICATION

1. **Jiajun Zhu**, Yuehao Wang, Peihao Wang, Ruisi Cai, Siqi Miao, Liliang Ren, Pan Li, Jason D. Lee, Zhangyang Wang. Self Interpretable, Robust SSMs with Stochastic Selective Memory, *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026. Submitted.
2. **Jiajun Zhu**, Peihao Wang, Ruisi Cai, Jason D. Lee, Pan Li, Zhangyang Wang. Rethinking Addressing in Language Models via Contextualized Equivariant Positional Encoding, *The Forty-Second International Conference on Machine Learning (ICML)*, 2025.
3. **Jiajun Zhu**, Siqi Miao, Rex Ying, Pan Li. Towards Understanding Sensitive and Decisive Patterns in Explainable AI: A Case Study of Model Interpretation in Geometric Deep Learning, *Nature Machine Intelligence*, 2025.
4. Peihao Wang, Ruisi Cai, Yuehao Wang, **Jiajun Zhu**, Pragya Srivastava, Zhangyang Wang, Pan Li, Understanding Bottlenecks of State Space Models through the Lens of Recency and Over-smoothing, *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

SELECTED RESEARCH

Improved, Robust and Interpretable Mamba with Stochastic Memory

Jan. 2025 - Jun. 2025

Supervised by Prof. Zhangyang (Atlas) Wang

University of Texas at Austin

- Proposed principled framework of IB-gated SSM with stochastic selection and two implemented variants to enhance generalization, robustness and providing interpretability.
- Achieved +1.38% absolute performance gain on 11 language benchmarks, while ensuring +28% higher interpretation fidelity and +40% accuracy improvement under adversarial attack through two model variants.

Long-Context Transformer with Equivariant Position Encoding

Jun. 2024 - Sept. 2024

Supervised by Prof. Zhangyang (Atlas) Wang

University of Texas at Austin

- Proposed principles of permutation invariance and orthogonal equivariance to enhance the generalization of learnable positional embeddings, and designed an enhanced Transformer in alignment with principles.
- Demonstrated +20% gains on long-context benchmark (Scrolls) and +22% on (position-sensitive) arithmetic reasoning with only +10% overhead, effective in both pretraining and finetuning via a fused-kernel implementation.

EXPERIENCE

Personalized Diffusion Models via VLM-based Reward Modeling

Oct. 2025 - Mar. 2026 (Est.)

Supervised by Dr. Neil Shah & Prof. Atlas Wang

Snap Inc.

- Proposed a method to personalize diffusion models via reinforcement finetuning (GRPO) with a VLM-based generative reward model conditioned on user embeddings.

Interactive Reasoning of Visual Language Models

Dec. 2023 - Apr. 2024

Supervised by Prof. Yaochu Jin

Westlake University

- Proposed a paradigm for interacting with VLMs via visual referencing, using “click-and-segment” actions to enhance interactivity and reference accuracy.
- Fine-tuned VLMs on a custom multimodal dataset, boosting image segmentation performance by 12% through improved reasoning ability.

SKILLS

- Programming languages: Python, C/C++, CUDA.
- Software & Frameworks: LaTeX, Git, PyTorch, PyTorch Geometric, Transformers, Triton.